

Functional Requirement Document for Asset Management Tool

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TECHSTAUNCH

1. Document purpose

This FRD defines the complete functional requirements for the Asset Management System. It covers all screens, data models, form fields, business rules, role-based permissions, workflow logic, and status definitions. It is intended to be the single source of truth for the engineering team and must clarify implementation requirements without ambiguity.

1.1 Project Scope

In Scope

- Asset Register (create, view, edit, bulk upload)
- Asset Listing (card/browse view)
- Asset Detail View
- Deployment Board
- Transfer Workflow
- Compliance Overview
- Reports
- Admin (User Management, Location, Categories, Role Permissions, Notification Settings, System Settings)
- Dashboard

Future Scope

- Full native implementation of advanced statistical tools
- Direct real-time integration with manufacturing equipment
- ERP/MES/PLM system integrations
- Mobile app
- Advanced AI/ML features
- Hardware procurement or on-premise deployment (cloud-only)
- Supplier onboarding portal or e-commerce features.

2. User Roles & Permissions

2.1 Role Definitions

Role	Description
IT Admin	Full system access. Can create, edit, delete all records, manage users, approve transfers, configure system settings.
IT Ops	Can view all assets, update service/compliance fields, initiate transfers, log maintenance. Cannot manage users or system config.
HR Department	Can view asset listing, request transfers for onboarding purposes. Cannot edit asset data.
Engineering Team	Can view assigned assets, request transfers, view deployment boards. Read-only access to others' assets.
Finance	Can view financial fields (purchase cost, depreciation, AMC). Cannot edit asset records.
Viewer/Read-only	Can view asset listing and basic details only. No write access.

2.2 Permission Matrix

Feature/Action	IT Admin	IT Ops	HR Dept	Engineering	Finance	Viewer
View Asset Register	✓	✓	✓	✓	✓	✓
Add Asset	✓	✓	✗	✗	✗	✗
Edit Asset	✓	✓	✗	✗	✗	✗
Delete Asset	✓	✗	✗	✗	✗	✗
Bulk Upload	✓	✓	✗	✗	✗	✗
Export Asset Data	✓	✓	✗	✗	✓	✗
View Financial Fields	✓	✗	✗	✗	✓	✗
View Compliance	✓	✓	✗	✗	✗	✗
Update Compliance	✓	✓	✗	✗	✗	✗
Request Transfer	✓	✓	✓	✓	✗	✗
Approve Transfer	✓	✗	✗	✗	✗	✗
View Deployment Board	✓	✓	✓	✓	✗	✗
Mark Available/Reassign	✓	✓	✗	✗	✗	✗
View Reports	✓	✓	✗	✗	✓	✗
Manage Users	✓	✗	✗	✗	✗	✗
Admin Configuration	✓	✗	✗	✗	✗	✗

3. Data Model & Field Definitions

3.1 Asset Record : Complete Field List

Section A: Basic Information (always visible to all roles with access)

Field Name	Field Type	Required	Editable	Validation Rules	Notes
Asset ID	Auto-generated text	System	No	Example format: AST-XXX (3-digit zero-padded sequential)	Generated on creation, immutable
Asset Type	Dropdown	Yes	Yes	Must match predefined category list	e.g., Laptop, Desktop, Server, Network Device, Storage, Peripheral
Model	Text (free)	Yes	Yes	Max 100 chars	e.g., "Dell Latitude 5540"
Configuration	Text (free)	Yes	Yes	Max 200 chars	e.g., "i7-1365U, 16GB RAM, 512GB SSD"
Serial Number	Text	Yes	No after creation	Must be unique across all assets; alphanumeric	Immutable post-creation
Vendor	Dropdown + freetext	Yes	Yes	Must match Vendor Master or allow manual entry	Linked to Vendor Master
Asset Photo	File upload	No	Yes	JPG/PNG, max 5MB	Optional; shown as thumbnail in card view

Section B: Assignment Info

Field Name	Field Type	Required	Editable	Validation Rules	Notes
Assigned To	Text / User lookup	No	Yes	Must be a named person or "Unassigned"	Free text (not restricted to system users)
Department	Dropdown	No	Yes	Predefined dept list from Admin	e.g., HR, Engineering, Finance, IT Ops
Location	Dropdown	Yes	Yes	Must match Location Master from Admin	e.g., HQ - Floor 1, Data Centre A, Warehouse
Status	Dropdown (controlled)	Yes	Conditional	See Status Rules (Section 4)	Not freely editable, governed by workflow
Condition	Dropdown	Yes	Yes	Excellent / Good / Fair / Poor	Manual assessment field

Section C: Financial Info (visible only to IT Admin and Finance roles)

Field Name	Field Type	Required	Editable	Validation Rules	Notes
Purchase Date	Date picker	Yes	Yes	Cannot be future date	
Purchase Cost	Currency (GBP)	Yes	Yes	Numeric, positive value	Stored in GBP (£)
Warranty Expiry	Date picker	Yes	Yes	Must be after Purchase Date	
AMC Expiry	Date picker	No	Yes	Must be after Purchase Date	Annual Maintenance Contract expiry
Depreciation Rate	Number (%)	No	Yes	0-100	Optional; used in Depreciation Report
Current Value	Calculated (read-only)	No	No	Auto-calculated from cost + depreciation rate + age	Display only

Section D: Compliance Info

Field Name	Field Type	Required	Editable	Validation Rules	Notes
Compliance Status	Dropdown (auto + manual)	System	Partially	See Compliance Rules (Section 6)	Auto-set by rules; manually overridable by IT Admin/IT Ops
Last Service Date	Date picker	No	Yes	Cannot be future date	
Next Service Due	Date picker	No	Yes	Must be after Last Service Date	Used for compliance auto-calculation
Service Notes	Textarea	No	Yes	Max 500 chars	

Section E: Additional / Metadata

Field Name	Field Type	Required	Editable	Validation Rules	Notes
Notes	Textarea	No	Yes	Max 1000 chars	General notes field
Spare Parts Info	Textarea	No	Yes	Max 500 chars	
Tags	Multi-select tag input	No	Yes	Max 10 tags, 30 chars each	Free-form or from tag library
Documents	File upload (multi)	No	Yes	PDF/JPG/PNG/DOCX, max 10MB each, max 5 files	Invoices, warranties, service reports
Created By	Text (system)	System	No	Auto-populated	Name of user who created record
Created At	Datetime (system)	System	No	Auto-populated	
Last Updated By	Text (system)	System	No	Auto-populated	
Last Updated At	Datetime (system)	System	No	Auto-populated	

4. Asset Status Rules

4.1 Status Values

Status	Badge Color	Description
In Use	Blue	Asset is actively deployed and assigned to a person
Available	Green	Asset is unassigned and ready for deployment
Idle	Yellow/Orange	Asset is unassigned and not in active use; may need assessment
Under Maintenance	Red	Asset is with service team; not available for assignment
Retired	Grey	Asset has been decommissioned; read-only record
In Transfer	Purple/Teal	Asset is mid-transfer; status locked until transfer closes

4.2 Status Transition Rules

Status cannot be changed freely via a dropdown. Transitions are triggered by specific actions:

From Status	Trigger Action	New Status	Who Can Trigger
Any	Asset first created	Available (default)	IT Admin, IT Ops
Available	"Assign to user" action	In Use	IT Admin, IT Ops
In Use	"Mark Available for Reassignment" button	Available	IT Admin, IT Ops
In Use / Available	"Send for Service" action	Under Maintenance	IT Admin, IT Ops
Under Maintenance	"Service Completed" action	Available	IT Admin, IT Ops
Available / Idle	Transfer Approved	In Transfer	System (auto)
In Transfer	Transfer Closed	In Use (at new location)	System (auto)

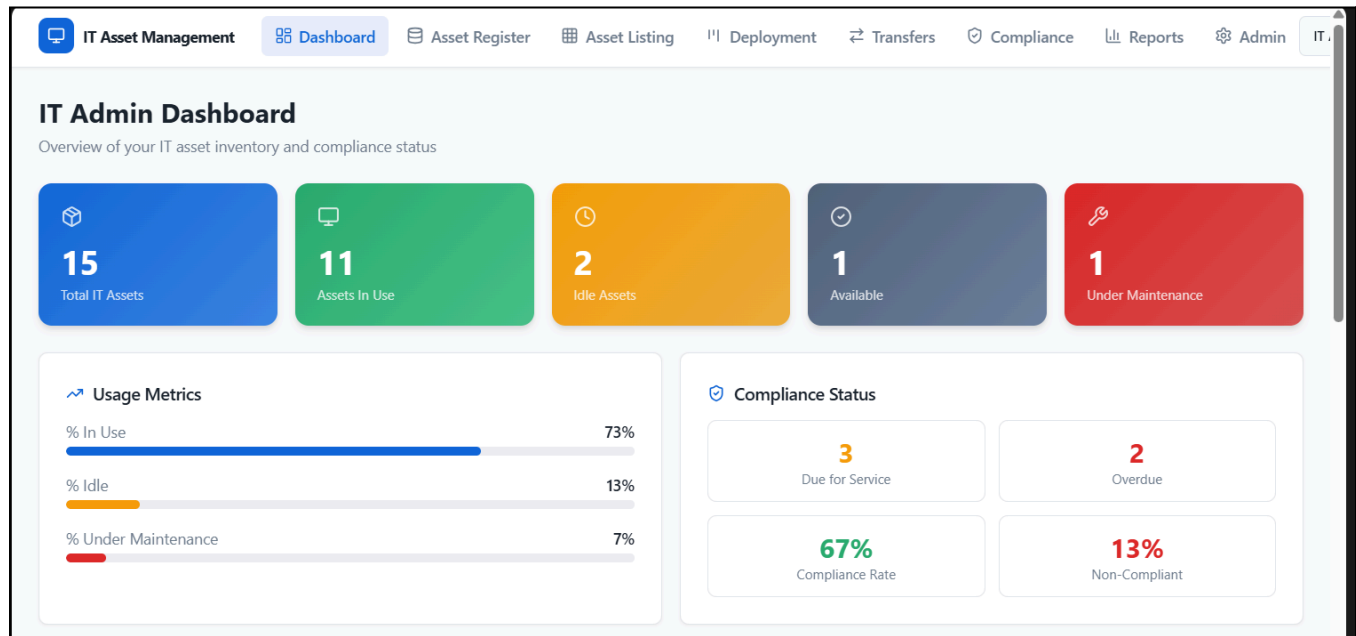
Any	Admin marks as Retired	Retired	IT Admin only
In Use -> not used for 90 days	Automated rule (optional config)	Idle	System (auto)

Rule: An asset with status **In Transfer** cannot be edited, transferred again, or have its status manually changed until the transfer is closed or rejected.

Rule: An asset with status **Retired** is read-only. No actions can be performed on it. It remains visible in the Asset Register filtered by "Retired" status.

5. Screens & Functional Specifications

5.1 Dashboard



Access: All roles (content varies by role)

Summary Cards (top row) : always visible:

- Total IT Assets
- Assets In Use
- Idle Assets
- Available
- Under Maintenance

Rule: Each card count is computed in real-time from the Asset Register. Clicking a card navigates to the Asset Register filtered by that status.

Usage Metrics Panel:

- $\% \text{ In Use} = (\text{Assets In Use} / \text{Total}) \times 100$
- $\% \text{ Idle} = (\text{Idle Assets} / \text{Total}) \times 100$
- $\% \text{ Under Maintenance} = (\text{Under Maintenance} / \text{Total}) \times 100$
- Displayed as labeled progress bars

Compliance Status Panel:

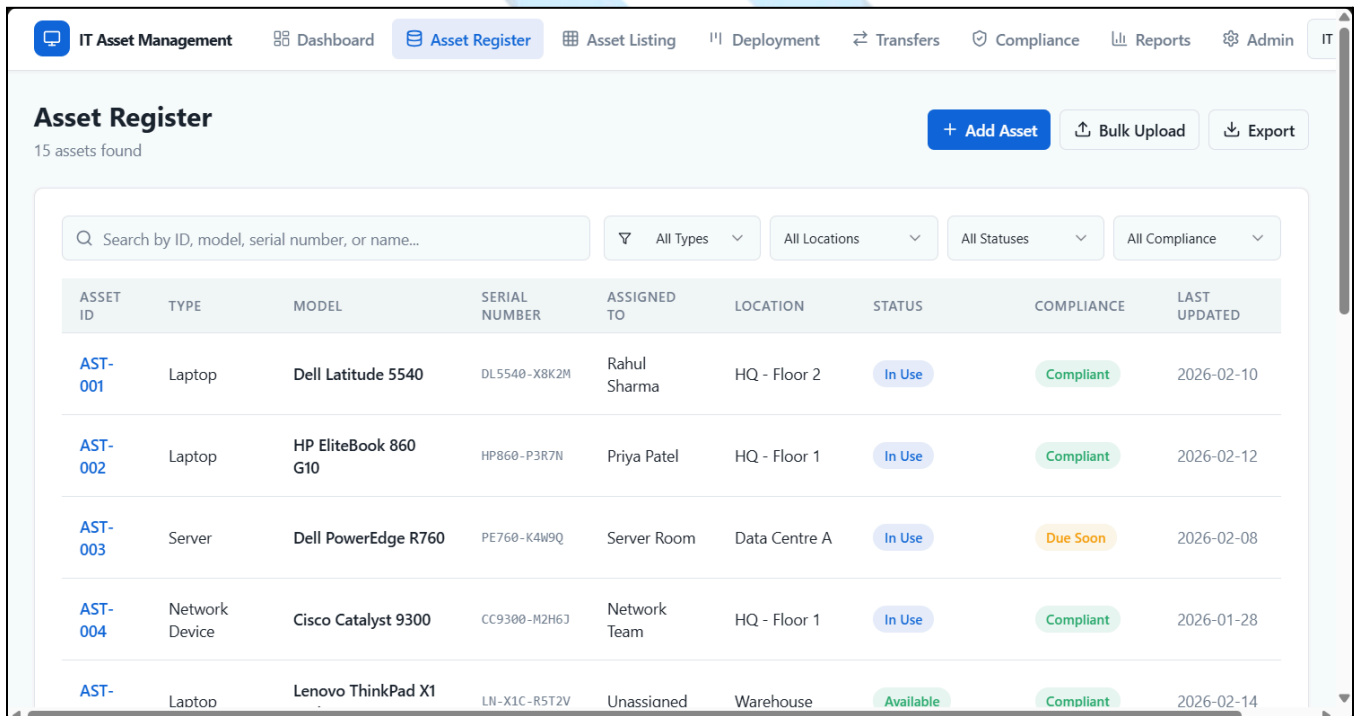
- Due for Service (count : assets where Next Service Due is within 30 days and status is not Overdue)
- Overdue (count : assets where Next Service Due has passed)
- Compliance Rate (% of assets in Compliant status)
- Non-Compliant (% of assets in Overdue or Non-Compliant status)

Chart Widgets (below the panels):

- Assets by Location : Bar chart; X-axis = location names; Y-axis = count of assets
- Asset Status Distribution : Donut/Pie chart using status colors
- Compliance Trend : Line chart; X-axis = last 6 months; Y-axis = compliance rate %

Role Note: Finance role sees Dashboard without the Compliance panel. The viewer sees summary cards only.

5.2 Asset Register



The screenshot shows the 'Asset Register' page in the IT Asset Management system. The page has a top navigation bar with links to Dashboard, Asset Register (active), Asset Listing, Deployment, Transfers, Compliance, Reports, and Admin. Below the navigation bar, the 'Asset Register' section displays '15 assets found' and includes buttons for '+ Add Asset', 'Bulk Upload', and 'Export'. A search bar and filter dropdowns (All Types, All Locations, All Statuses, All Compliance) are present. The main table lists assets with columns: ASSET ID, TYPE, MODEL, SERIAL NUMBER, ASSIGNED TO, LOCATION, STATUS, COMPLIANCE, and LAST UPDATED. The table shows five rows of asset data.

ASSET ID	TYPE	MODEL	SERIAL NUMBER	ASSIGNED TO	LOCATION	STATUS	COMPLIANCE	LAST UPDATED
AST-001	Laptop	Dell Latitude 5540	DL5540-X8K2M	Rahul Sharma	HQ - Floor 2	In Use	Compliant	2026-02-10
AST-002	Laptop	HP EliteBook 860 G10	HP860-P3R7N	Priya Patel	HQ - Floor 1	In Use	Compliant	2026-02-12
AST-003	Server	Dell PowerEdge R760	PE760-K4W9Q	Server Room	Data Centre A	In Use	Due Soon	2026-02-08
AST-004	Network Device	Cisco Catalyst 9300	CC9300-M2H6J	Network Team	HQ - Floor 1	In Use	Compliant	2026-01-28
AST-005	Laptop	Lenovo ThinkPad X1	LN-X1C-R5T2V	Unassigned	Warehouse	Available	Compliant	2026-02-14

Access: IT Admin, IT Ops, HR Dept, Engineering, Finance, Viewer (column-level restrictions apply)

5.2.1 Page-Level Actions

Action	Roles	Behavior
Add Asset	IT Admin, IT Ops	Opens asset creation form (inline drawer or modal). All sections from Section 3 are presented. Asset ID is generated on save.
Bulk Upload	IT Admin, IT Ops	See Section 5.2.4
Export	IT Admin, IT Ops, Finance	Exports current filtered result set. Financial columns included only for IT Admin and Finance. Format: Excel (.xlsx).

5.2.2 Search & Filtering

- Text search: Full-text search across Asset ID, Model, Serial Number, and Assigned To fields. Debounced at 300ms.
- Type filter: Multi-select. Values populated dynamically from Asset Categories master.
- Location filter: Multi-select. Values populated dynamically from Location Master.
- Status filter: Multi-select. Fixed values: In Use, Available, Idle, Under Maintenance, In Transfer, Retired.
- Compliance filter: Multi-select. Fixed values: Compliant, Due Soon, Overdue, Non-Compliant.

All active filters are combinative (AND logic). The result count updates dynamically.

5.2.3 Asset Table

Default sort: Last Updated descending.

Column	Sortable	Notes
Asset ID	Yes	Rendered as a hyperlink to Asset Detail page
Type	Yes	
Model	Yes	
Serial Number	No	
Assigned To	Yes	Displays "Unassigned" when no assignment exists
Location	Yes	
Status	Yes	Rendered as a color-coded badge
Compliance	Yes	Rendered as a color-coded badge. Hidden for roles without compliance access.
Last Updated	Yes	Date format per system date format setting

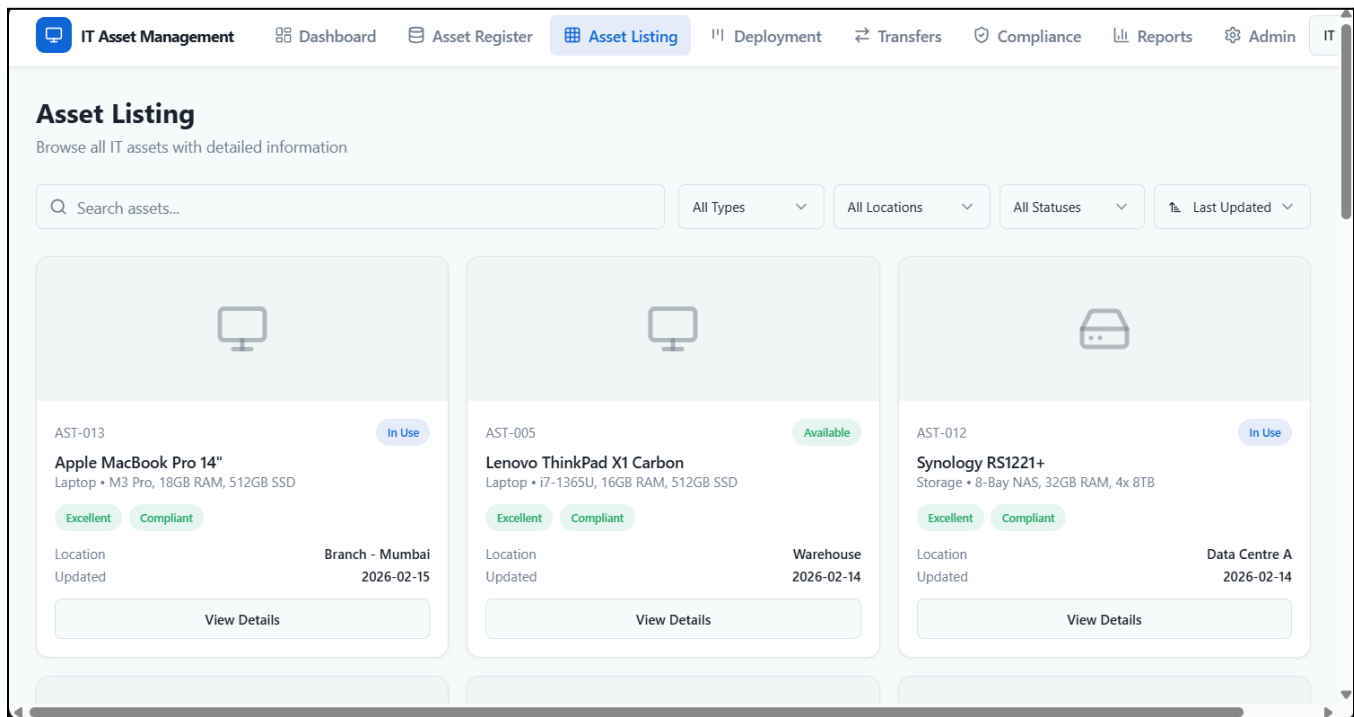
Financial columns are not displayed in the table under any role. They are only accessible on the Asset Detail page.

Pagination: Default 20 rows per page. Options: 20, 50, 100. Total count and current page indicator shown.

5.2.4 Bulk Upload

- Accepted file formats: .xlsx, .csv
- A downloadable blank template is available; column headers match all required fields from Section 3
- On file upload: all rows are validated before any records are created
- Validation checks: required fields present, serial number uniqueness, valid dropdown values (Type, Location, Vendor), date format correctness
- Validation results are displayed as a row-by-row preview with error highlighting per cell
- If any row contains an error, the entire upload is blocked; partial imports are not permitted
- On successful validation and user confirmation: all records are created, Asset IDs generated in row order

5.3 Asset Listing (Card View)



Access: IT Admin, IT Ops, HR Dept, Engineering, Viewer

A card-based alternate view of the asset register. Functionally equivalent to the Asset Register in terms of data shown, but presented in a visual grid layout rather than a table.

5.3.1 Card Data Fields

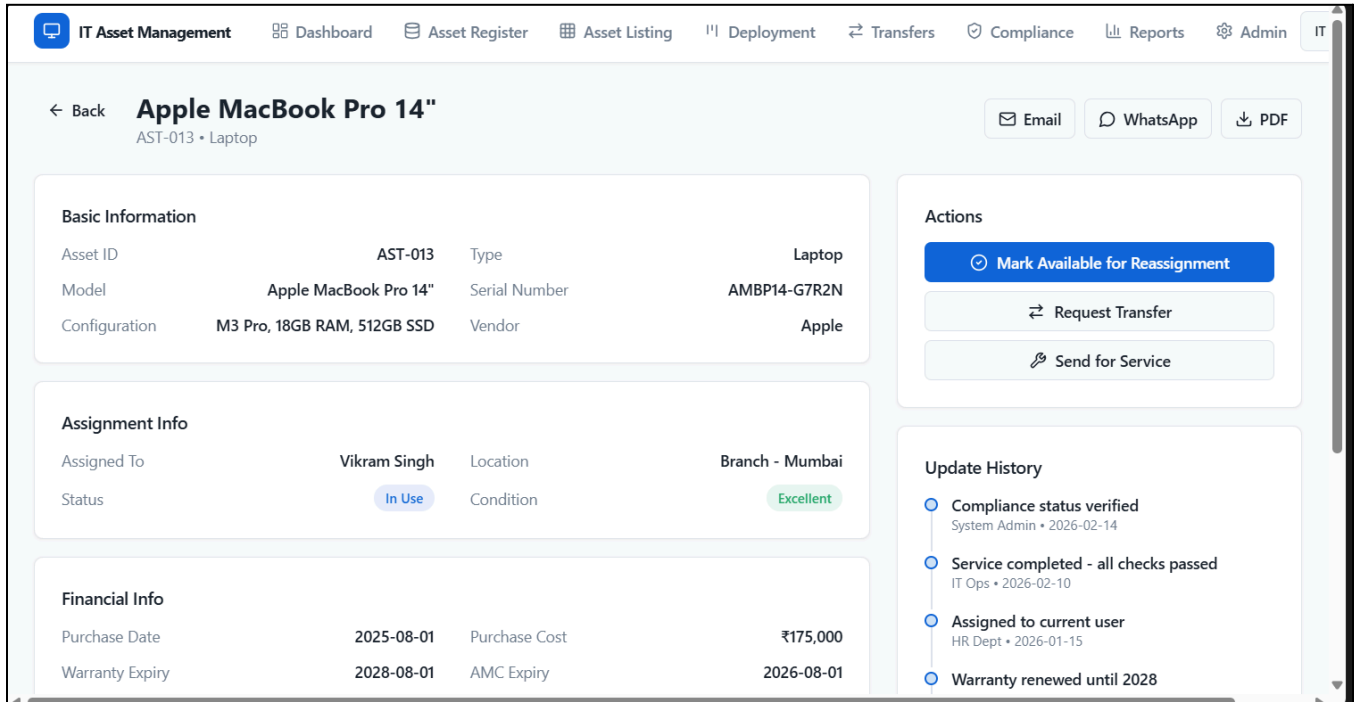
Each card displays: Asset ID, Status badge, Asset type icon (derived from asset type), Model name, Type and Configuration (combined sub-label), Condition badge, Compliance badge (hidden for roles without compliance access), Location, Last Updated date, and a "View Details" action that navigates to the Asset Detail page.

Financial information is not displayed on cards under any role.

5.3.2 Search, Filter & Sort

- Text search: same scope as Asset Register (Asset ID, Model, Serial Number, Assigned To)
- Type filter, Location filter, Status filter: same behavior as Asset Register
- Sort control: Last Updated (default), Model A-Z, Asset ID ascending

5.4 Asset Detail Page



The screenshot shows the 'Apple MacBook Pro 14"' asset detail page. The page header includes navigation links: IT Asset Management, Dashboard, Asset Register, Asset Listing, Deployment, Transfers, Compliance, Reports, and Admin. The asset is identified as 'AST-013 • Laptop'. Action buttons for Email, WhatsApp, and PDF are available. The page is divided into three main sections: Basic Information, Assignment Info, and Financial Info, each containing a table of details. An 'Actions' sidebar on the right provides options to mark the asset as available for reassignment, request a transfer, or send it for service. An 'Update History' section on the right lists recent events such as compliance verification, service completion, and warranty renewal.

Basic Information

Asset ID	AST-013	Type	Laptop
Model	Apple MacBook Pro 14"	Serial Number	AMBP14-G7R2N
Configuration	M3 Pro, 18GB RAM, 512GB SSD	Vendor	Apple

Assignment Info

Assigned To	Vikram Singh	Location	Branch - Mumbai
Status	In Use	Condition	Excellent

Financial Info

Purchase Date	2025-08-01	Purchase Cost	₹175,000
Warranty Expiry	2028-08-01	AMC Expiry	2026-08-01

Actions

- Mark Available for Reassignment
- Request Transfer
- Send for Service

Update History

- Compliance status verified (System Admin • 2026-02-14)
- Service completed - all checks passed (IT Ops • 2026-02-10)
- Assigned to current user (HR Dept • 2026-01-15)
- Warranty renewed until 2028

Access: All roles (field-level visibility per Section 2.1)

The Asset Detail page is the single canonical view of a complete asset record.

5.4.1 Page Actions

Action	Visible When	Available To	Behavior
Mark Available for Reassignment	Status = In Use	IT Admin, IT Ops	Transitions asset status from In Use → Available. Clears Assigned To field. Logs event in Update History.
Request Transfer	Status = In Use or Available	IT Admin, IT Ops, HR Dept, Engineering	Opens transfer request form (see Section 5.6).
Send for Service	Status ≠ Under Maintenance AND ≠ Retired AND ≠ In Transfer	IT Admin, IT Ops	Transition status to Under Maintenance. Prompts for Service Notes (optional). Logs event in Update History.

Service Completed	Status = Under Maintenance	IT Admin, IT Ops	Transitions status to Available. Prompts for Service Notes (optional). Logs event.
Edit Asset	Any status except Retired and In Transfer	IT Admin, IT Ops	Opens all editable fields in edit mode. Non-mutable fields (Asset ID, Serial Number) remain locked.
Retire Asset	Any status except Retired	IT Admin only	Requires confirmation dialog with mandatory reason text. Irreversible. Sets status to Retired, locks record. Logs event.

5.4.2 Share Actions

Three sharing options are available on the Asset Detail page:

- **Email:** Opens the system default mail client with a pre-composed plain-text body containing Basic Information and Assignment Information only.
- **WhatsApp:** Opens WhatsApp (web or native) with a pre-formatted plain-text message containing Basic Information and Assignment Information only.
- **Download PDF:** Generates and downloads a formatted PDF containing Basic Information, Assignment Information, Compliance Information (if user role has access), and the last 5 Update History entries.

Financial information is excluded from all share and export actions on this page regardless of role.

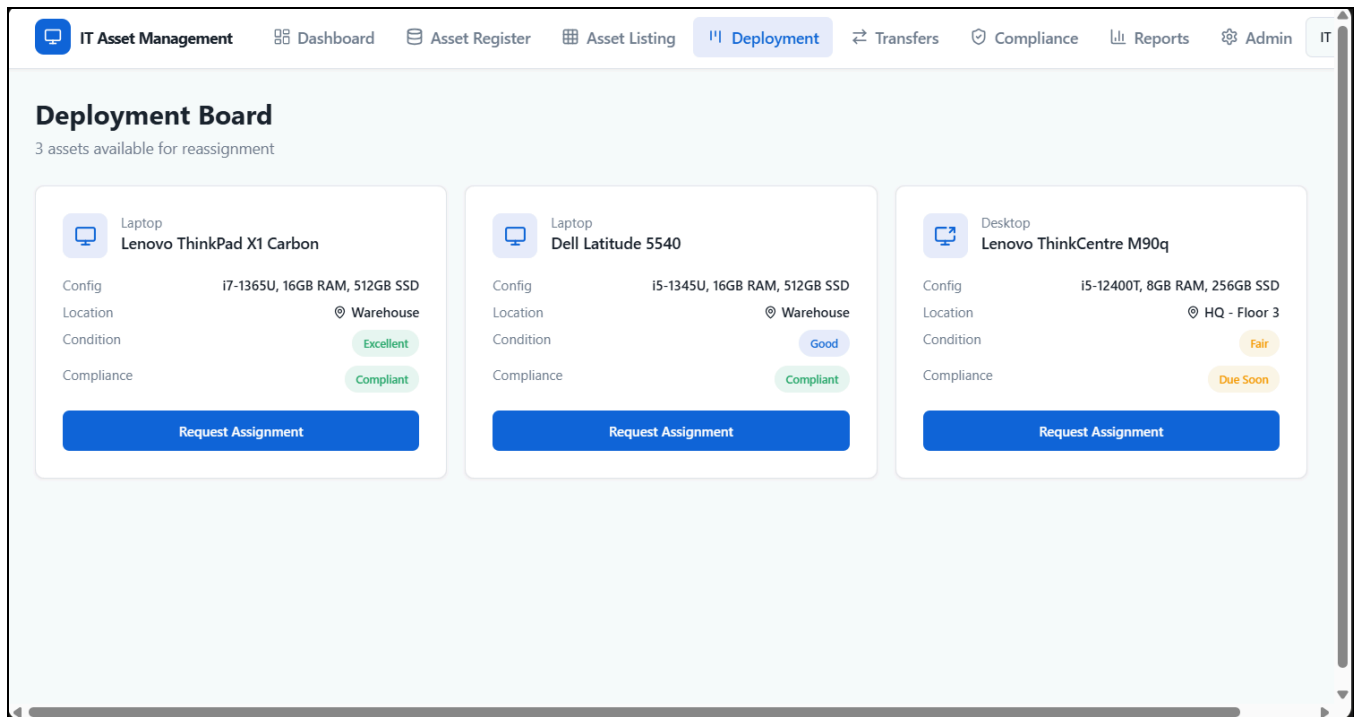
5.4.3 Update History

A chronological, append-only log of all changes to the asset record. Each entry contains:

- Description of the event (e.g., "Status changed from In Use to Under Maintenance")
- Name and role of the user who triggered the event
- UTC timestamp

History entries are system-generated and cannot be edited or deleted. Displayed in reverse-chronological order (most recent first). Visible to: IT Admin, IT Ops

5.5 Deployment Board



Access: IT Admin, IT Ops, HR Dept, Engineering

The Deployment Board displays all assets that are currently available for reassignment, providing a single place for requesters to identify and claim available assets before new procurement is initiated.

5.5.1 Board Rules

- Only assets with status = **Available** are displayed on the Deployment Board
- Assets with any other status (In Use, Idle, Under Maintenance, In Transfer, Retired) are excluded
- The page header shows a live count of available assets

5.5.2 Card Data Fields

Each card displays: Asset Type, Model name, Configuration, Location, Condition badge, Compliance badge (visible to IT Admin and IT Ops only), and a "Request Assignment" action button

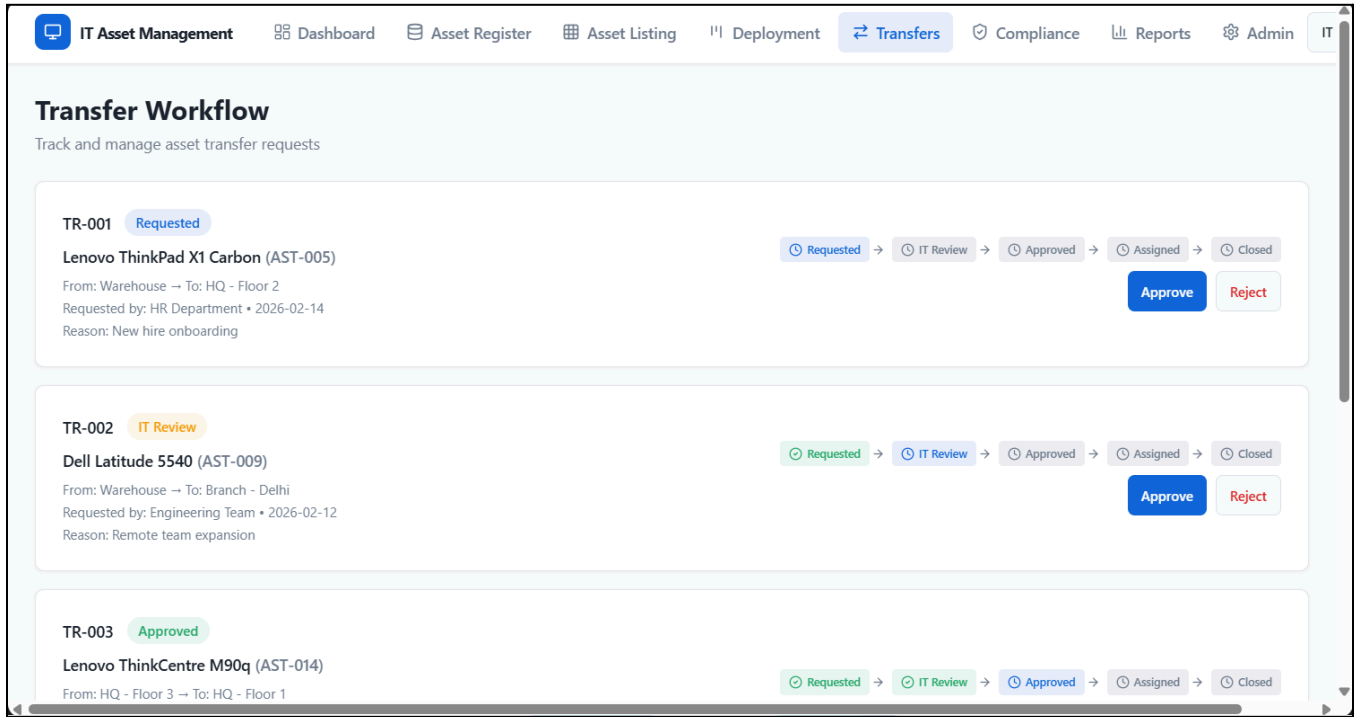
5.5.3 Request Assignment Flow

- User clicks "Request Assignment" on an asset card
- A form is presented with the following fields:

Field	Type	Required	Notes
Requested For	Text	Yes	Name of the person who will receive the asset
Department	Dropdown	Yes	Must match department master
Target Location	Dropdown	Yes	Must match Location Master
Reason	Textarea	Yes	Max 300 characters

- On submission: a Transfer Request record is created with status = Requested. The asset status changes to **In Transfer**. An in-app and email notification is sent to all IT Admin users.
- The asset is removed from the Deployment Board immediately upon submission.

5.6 Transfer Workflow



Transfer Workflow
Track and manage asset transfer requests

TR-001 Requested

Lenovo ThinkPad X1 Carbon (AST-005)

From: Warehouse → To: HQ - Floor 2

Requested by: HR Department • 2026-02-14

Reason: New hire onboarding

Requested → IT Review → Approved → Assigned → Closed

Approve Reject

TR-002 IT Review

Dell Latitude 5540 (AST-009)

From: Warehouse → To: Branch - Delhi

Requested by: Engineering Team • 2026-02-12

Reason: Remote team expansion

Requested → IT Review → Approved → Assigned → Closed

Approve Reject

TR-003 Approved

Lenovo ThinkCentre M90q (AST-014)

From: HQ - Floor 3 → To: HQ - Floor 1

Requested → IT Review → Approved → Assigned → Closed

Access: IT Admin (full access), IT Ops (view all, limited actions), HR Dept and Engineering (view and manage own requests only)

5.6.1 Transfer Record Fields

Each transfer record stores:

Field	Type	Notes
Transfer ID	Auto-generated string	Format: TR-001, sequential, never reused
Asset ID	Reference	Linked asset
Asset Model	Denormalized string	Snapshot at time of request
From Location	String	Location at time of request
To Location	String	Target location from request form

Requested By	String	Name and role/department of requester
Requested For	String	Intended recipient
Request Date	Date	Date request was submitted
Reason	Text	From request form
Status	Controlled	See Section 5.6.2
Status History	Array	Log of each status change with user and timestamp

5.6.2 Transfer Status Definitions

Status	Description	Who Sets It
Requested	Transfer request submitted; awaiting IT review	System (automatic on form submit)
IT Review	IT Admin has acknowledged the request and is reviewing it	IT Admin (manual action)
Approved	Request approved; asset to be physically moved	IT Admin (manual action)
Assigned	Asset has been physically moved and assigned to new user at new location	IT Admin or IT Ops (manual action)
Closed	Transfer complete; record locked	System (automatic on Assigned confirmation)
Rejected	Request denied at any stage	IT Admin (manual action)

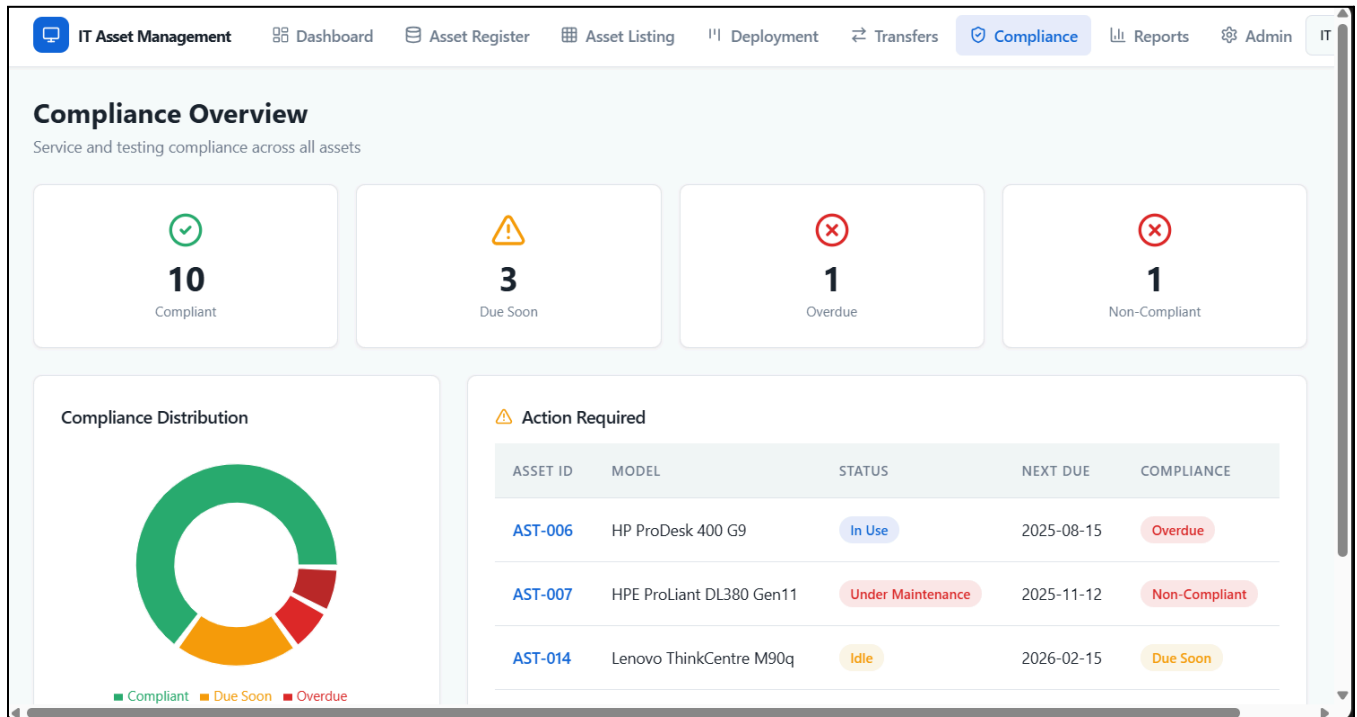
5.6.3 Transfer Workflow Rules

- When a transfer is created, the linked asset's status is immediately set to In Transfer and the asset is locked.
- When a transfer reaches Closed status, the system automatically updates the asset record: Location is set to the transfer's To Location, Assigned To is set to the Requested For value, and Status is set to In Use.
- When a transfer is Rejected, the asset status is automatically reverted to its status prior to the transfer being initiated (Available or In Use as applicable).
- An IT Admin can reject a transfer at any stage before it reaches Closed.
- HR Dept and Engineering users can view all transfers but can only take action on (cancel, view details of) transfers they personally initiated. They cannot Approve or Reject transfers.
- Only one active (non-Closed, non-Rejected) transfer may exist per asset at any time.

5.6.4 Filtering & Search

- Filter by Status: All, Requested, IT Review, Approved, Assigned, Closed, Rejected
- Text search: by Asset ID, Transfer ID, or asset model name

5.7 Compliance Overview



Access: IT Admin, IT Ops

5.7.1 Summary Metrics

Four summary count cards are displayed:

Metric	Definition
Compliant	Count of non-Retired assets where compliance status = Compliant
Due Soon	Count of non-Retired assets where compliance status = Due Soon
Overdue	Count of non-Retired assets where compliance status = Overdue
Non-Compliant	Count of non-Retired assets where compliance status = Non-Compliant

5.7.2 Action Required Table

- Displays all assets with compliance status of Due Soon, Overdue, or Non-Compliant.
- Columns: Asset ID (link to detail), Model, Status badge, Next Service Due date, Compliance badge.
- Default sort: Next Service Due ascending (most urgent first).

5.7.3 Compliance Distribution Chart

A chart showing the proportion of assets in each compliance state. Data derived from the same counts as the summary metrics above.



6. Compliance Rules Engine

Compliance status is calculated automatically by the system. The rules engine runs as a scheduled daily job (recommended: midnight server time) and also recalculates on any individual asset page load.

6.1 Automated Rules

Condition	Resulting Compliance Status
Next Service Due is more than 30 days in the future	Compliant
Next Service Due is within 30 days from today (inclusive)	Due Soon
Next Service Due date is in the past	Overdue
AMC Expiry date is in the past	Non-Compliant (flagged independently; additive with service compliance)
Next Service Due is not set and asset has never been serviced	Compliant (default until a service date is configured)

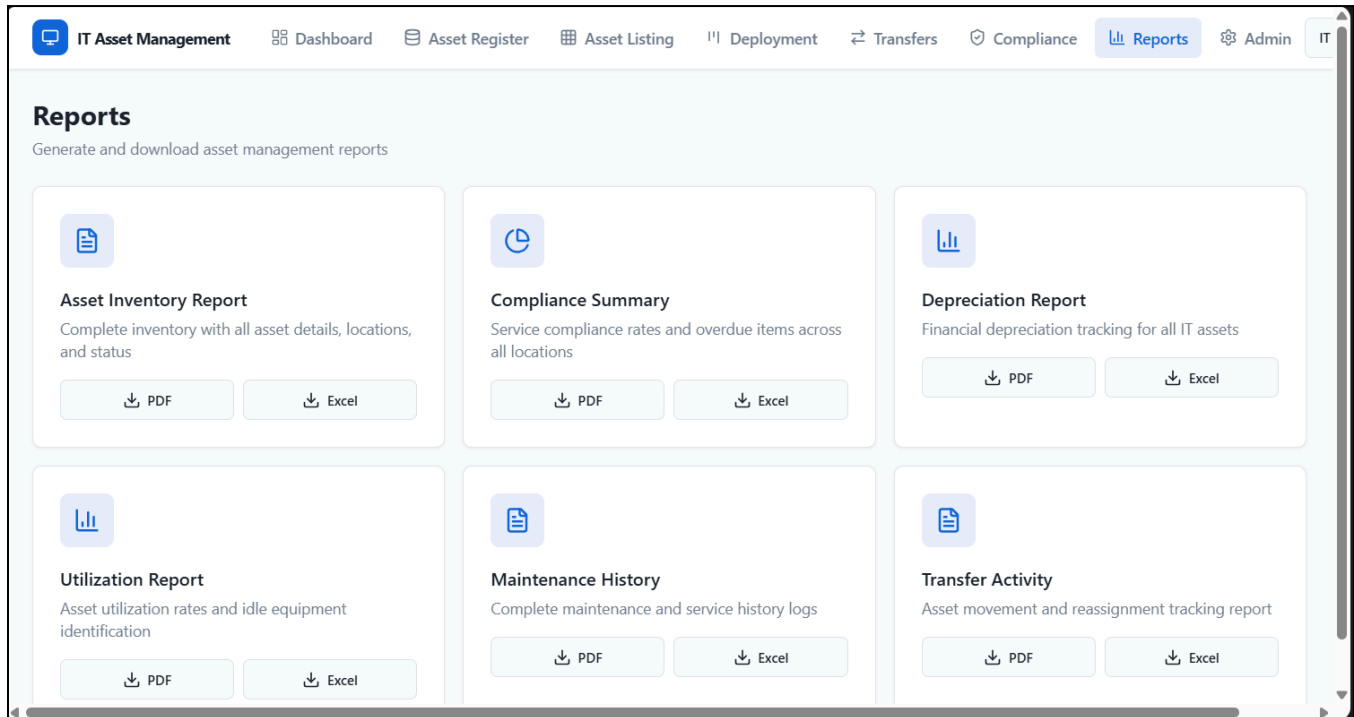
6.2 Manual Override

IT Admin and IT Ops may manually override the compliance status of any asset. A manual override requires a reason note (max 200 characters) which is recorded in the Update History. A manually set status persists until the next scheduled rules engine run, at which point automated rules re-evaluate and may override the manual setting unless a hard-lock flag is set by IT Admin.

6.3 Retired Asset Exclusion

Assets with status = Retired are excluded from all compliance calculations, counts, and reports.

7. Reports Module



Access: IT Admin, IT Ops (all reports); Finance (Depreciation Report and Utilization Report only). All reports are generated on-demand. No scheduling or automated delivery is required in v1.

7.1 Available Reports

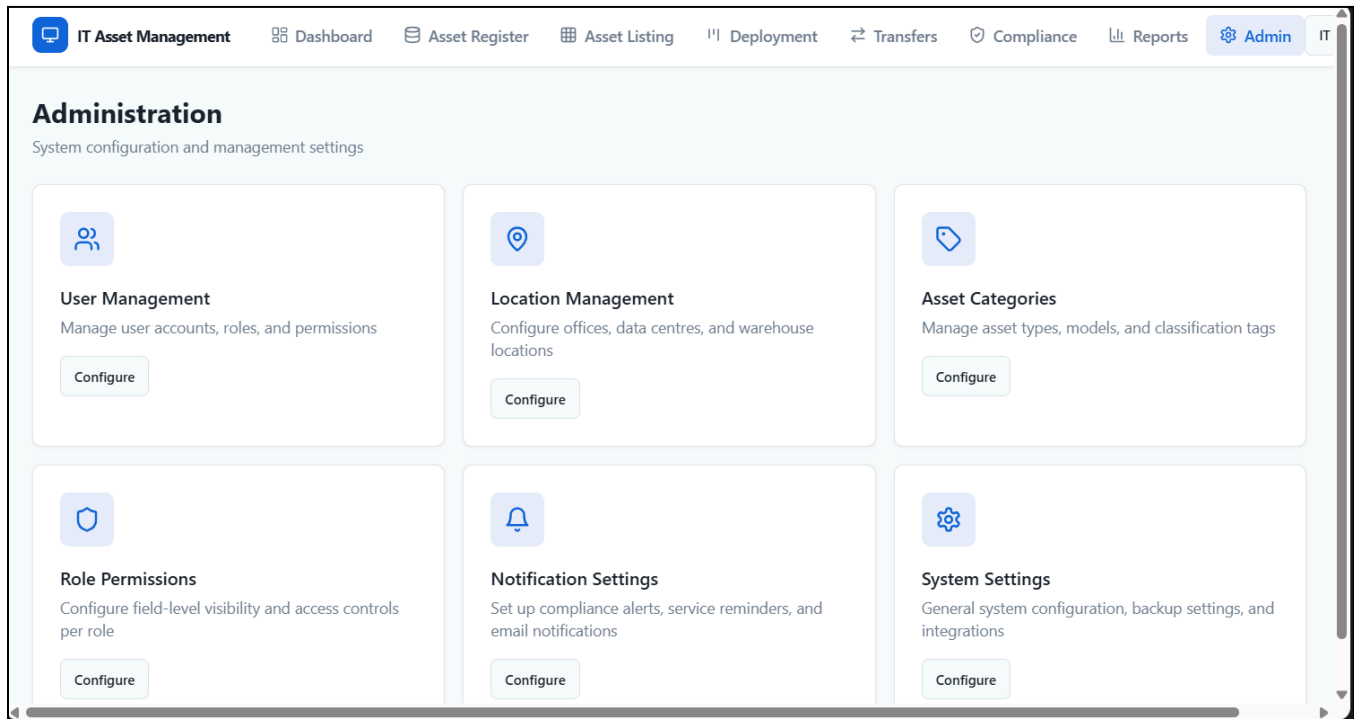
Report	Description	Available Formats	Financial Data Included
Asset Inventory Report	Complete list of all assets with all fields, current status, and location	PDF, Excel	Yes (IT Admin only)
Compliance Summary	Compliance rates, overdue counts, and action items broken down by location and asset type	PDF, Excel	No

Depreciation Report	Per-asset financial depreciation tracking; purchase cost, depreciation rate, current calculated value	PDF, Excel	Yes
Utilization Report	Asset utilization rates (% In Use, % Idle) by type and location; identifies underutilized equipment	PDF, Excel	No
Maintenance History	Full service log for all assets; filterable by asset ID and date range	PDF, Excel	No
Transfer Activity	All transfer records; asset movement and reassignment history	PDF, Excel	No

7.2 Report Generation Rules

- PDF reports must include: organization name (from System Settings), report title, date and time of generation, and page numbers.
- Excel reports must contain raw tabular data with column headers. No formulas are required in v1.
- Financial columns are included only when the requesting user's role has financial field access.
- Retired assets are excluded from all reports by default. An optional toggle to include Retired assets may be provided.
- Reports are generated synchronously for datasets under 1,000 records. For larger datasets, async generation with a download link notification is recommended.

8. Admin Module



Access: IT Admin only

8.1 User Management

Manages all system user accounts.

Field	Type	Required	Notes
Full Name	Text	Yes	Max 100 chars
Email Address	Email	Yes	Unique. Used as login identifier.
Role	Dropdown	Yes	Values from role list in Section 2.1
Department	Dropdown	No	From department master
Status	Toggle	Yes	Active / Inactive
Created Date	Date (system)	Auto	

Rules:

- A user cannot deactivate their own account.
- Deactivated users cannot log in. Their records are retained for audit trail integrity.
- Hard deletion of user accounts is not supported. Only deactivation (soft delete).
- Email address cannot be changed after account creation without IT Admin confirmation.

8.2 Location Management

Field	Type	Required	Notes
Location Name	Text	Yes	Max 100 chars. Must be unique.
Location Type	Dropdown	Yes	Values: Office, Data Centre, Warehouse, Branch
City	Text	No	Max 100 chars
Description	Textarea	No	Max 300 chars

Rules:

- A location cannot be deleted if any asset record references it.
- Location names are used in all dropdowns across the system; changes propagate immediately.

8.3 Asset Categories

Field	Type	Required	Notes
Category Name	Text	Yes	Max 100 chars. Must be unique.
Parent Category	Dropdown (self-referential)	No	Enables two-level hierarchy (e.g., Laptop → Ultrabook)
Description	Textarea	No	Max 300 chars

Rules:

- A category cannot be deleted if any asset record references it.
- Sub-categories are supported one level deep. A category with a parent cannot itself be a parent.

8.4 Role Permissions (Field-Level Configuration)

Provides a UI for IT Admin to configure which fields are visible and/or editable per role. The defaults are defined in [Section 2.1](#) of this document.

Rules:

- Core system fields (Asset ID, Serial Number, Status, Asset Type, Model, Location) cannot be hidden for any role.
- Custom fields (Tags, Notes, Spare Parts Info) can be toggled per role.
- Changes to role permissions take effect on the user's next page load (no session restart required).

8.5 Notification Settings

Configures which system events trigger notifications, and via which channels.

Trigger Event	Default Recipients	Default Channels
Transfer Request submitted	All IT Admin users	In-app, Email
Transfer Approved	Requesting user	In-app, Email
Transfer Rejected	Requesting user	In-app, Email
Asset compliance status changes to Overdue	IT Admin, IT Ops	In-app, Email
Next Service Due within 30 days	IT Admin, IT Ops	Email
AMC Expiry within 30 days	IT Admin	Email
Warranty Expiry within 30 days	IT Admin	Email

Each trigger can be independently enabled or disabled. Email notification bodies are system-defined in v1 and are not customizable by end users.

8.6 System Settings

Setting	Type	Default	Notes
Organization Name	Text	-	Used in PDF report headers
Organization Logo	File upload	-	PNG or JPG, max 2MB. Used in PDF headers.
Default Currency	Display only	GBP (£)	Hardcoded in v1. Field present for future extensibility.
Date Format	Dropdown	DD/MM/YYYY	Options: DD/MM/YYYY, YYYY-MM-DD
Auto-Idle Threshold	Integer (days)	90	Number of days after which an In Use asset with no record updates is automatically set to Idle. Configurable range: 30–180 days. Can be disabled (set to 0).
Data Export / Backup	Action button	-	Triggers a full data export of all asset records as an Excel file. Includes all fields. Available on-demand only.

9. System-Wide Business Rules

9.1 Asset ID Generation

- Format: AST- (EXAMPLE) prefix followed by a zero-padded sequential integer
- IDs are sequential and global (not per-category or per-location)
- An ID once assigned is never reused, even if the asset is Retired
- In bulk uploads, IDs are generated in the row order of the upload file

9.2 Audit Trail

Every create, update, status change, and workflow event on an asset record generates an immutable Update History entry. The entry captures: event description, user name, user role, and UTC timestamp. History entries cannot be edited or deleted by any role including IT Admin.

9.3 Soft Deletes Only

No hard deletions are permitted for any primary entity (assets, users, locations, categories, transfers). All deactivation or removal operations are soft deletes that set an `is_active` or equivalent flag to false while retaining the record in the database.

9.4 Export Content Rules

- All exports reflect the filtered state of the view at the time of export
- Financial columns are excluded from all exports for roles without financial field access, at the API level (not just front-end hiding)
- Export files are generated server-side and served as file downloads

9.5 Monetary Value Storage

All monetary values are stored as integers in the smallest currency unit (Penny for GBP). All display formatting (£ symbol, comma separation, decimal display) is handled in the presentation layer.

10. Non-Functional Requirements

Requirement	Specification
Database	Relational. All entities in structured tables with foreign key constraints.
File Storage	Cloud object storage (AWS S3 or equivalent) for asset photos and document attachments. Signed URLs for access.
Compliance Engine	Cron job scheduled daily at midnight (server time). Also triggered on individual asset record load for that specific asset.
Offline Mode	Not required in v1
Mobile App	Not required in v1. Web applications must be responsive down to tablet viewport (768px).
Concurrent Users	Must support up to 100 concurrent users without degradation in v1
Data Backup	Full database backup daily. Manual export available from System Settings.
Audit Log Retention	All Update History records are retained indefinitely. No automated purge.
Report Generation	Synchronous for $\leq 1,000$ asset records. Async with notification for larger datasets.
Browser Support	Latest two major versions of Chrome, Firefox, Safari, and Edge